

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0374345393s cost=1.41570093416 lag_gradient={'rotor':
-0.447274034409095, 'gimbal': -8.005418672407671e-07} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
7.984924186477807, 'finite_difference_l2_norm': 7.98492726697488, 'difference_l2_norm':
0.0007818443727450856, 'relative_error': 9.791502747667359e-05}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 7.984926576649946, 'finite_difference_l2_norm': 7.984929656676238,
'difference_l2_norm': 0.000781844211689216, 'relative_error': 9.791497800303255e-05},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.17554030173863, 'finite_difference_l2_norm': 17.175540220978167,
'difference_l2_norm': 7.092134352530405e-07, 'relative_error': 4.129205968450663e-08}]
smooth width=2: rotor=0.2s gimbal=0.0373614785s cost=1.41698239299 lag_gradient={'rotor':
-0.40878921360262604, 'gimbal': 5.818633143722662e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.858621093318575, 'finite_difference_l2_norm': 8.858666925424814,
'difference_l2_norm': 0.003075567195603299, 'relative_error': 0.00034718171723741736},
{'direction': 'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.858606591037159, 'finite_difference_l2_norm': 8.858652421238515,
'difference_l2_norm': 0.0030755671869557447, 'relative_error': 0.00034718228469852915},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.677881152810823, 'finite_difference_l2_norm': 18.677880511656927,
'difference_l2_norm': 5.279582558228024e-06, 'relative_error': 2.826649615678437e-07}]
smooth width=1: rotor=0.2s gimbal=0.0370789004s cost=1.41896500877 lag_gradient={'rotor':
-0.5404711913391944, 'gimbal': 0.02286610321069174} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
10.819479721997611, 'finite_difference_l2_norm': 10.820030460680568, 'difference_l2_norm':
0.010069750783404932, 'relative_error': 0.0009306582657043237}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.725587354933714, 'finite_difference_l2_norm': 10.726167695173112,
'difference_l2_norm': 0.010086880403996345, 'relative_error': 0.0009403992824516017},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 26.156050842934444, 'finite_difference_l2_norm': 25.91820652599986,
'difference_l2_norm': 2.146332389323925, 'relative_error': 0.08205873288030083}]
smooth width=0.5: rotor=0.2s gimbal=0.0363934178s cost=1.42081217044 lag_gradient={'rotor':
-0.873909674972295, 'gimbal': 0.017787688546570757} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
13.370612092602888, 'finite_difference_l2_norm': 13.373422078582804, 'difference_l2_norm':
0.03143145820038082, 'relative_error': 0.0023502928431996103}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 13.33255468923799, 'finite_difference_l2_norm': 13.335354326615752,
'difference_l2_norm': 0.03144432531535249, 'relative_error': 0.002357966990992765},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 38.71731300031717, 'finite_difference_l2_norm': 38.56694343557728,
'difference_l2_norm': 2.1104336667251045, 'relative_error': 0.05450878439595785}]

strict profile iterations: 1

iteration 1: best=(0.195002269745, 0.036393417803) cost=1.4206294313 rotor_range=(0.19000453949,
0.2) gimbal_range=(0.031394733874, 0.041392101732) expansions=[{'best_before_expansion':

```
{'rotor_delay_seconds': 0.195002269745, 'gimbal_delay_seconds': 0.036393417803, 'cost':  
1.4206294312964576}, 'added': [{'axis': 'rotor', 'delay_seconds': 0.19000453949}]}
```

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selected rotor lag 0.195002269745 s (39.0182 publish periods)  
selected gimbal lag 0.036393417803 s (7.2806 publish periods)  
selected objective cost 1.42062890573
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